

JAYDIP SINGH, PH.D.

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Research Profile](#)

PROFESSIONAL SUMMARY

Ph.D.-qualified Machine Learning Scientist with 10+ years of combined academic and industry experience in scientific computing, software engineering, and end-to-end AI/ML systems. Proven expertise in applying advanced machine learning, deep learning (CNNs, GNNs), and Generative AI techniques (LLMs, RAG) to solve complex problems across research and enterprise domains. Lead Engineer (GenAI) at Apexon, leading client-facing, enterprise-scale agentic and GenAI solutions. Experienced in building scalable data pipelines, automating training workflows, and deploying production ML systems on cloud platforms (AWS, Azure, GCP). Strong record of interdisciplinary collaboration, research-driven innovation, and delivery of impactful AI solutions, with a primary focus on agentic AI and multi-agent architectures.

Featured portfolio: github.com/JaydipSingh/portfolio (Rust + Python SLM framework, checkpointed training pipeline, and research-aligned ML engineering projects).

TECHNICAL SKILLS

- **Programming:** Python (NumPy, Pandas, PyTorch, Keras, TensorFlow, Scikit-learn, PySpark), C/C++, SQL, Bash/Shell.
- **Technology:** Machine Learning, Deep Learning, Generative AI, Agentic AI, LLMs, RAG (LangChain, MCP), CNNs, Monte Carlo Simulation, and Statistical Data Modeling.
- **Tools & Frameworks:** LangChain, LangGraph, LangSmith, OpenAI Agents SDK, CrewAI, AutoGen, MCP, FastAPI/RestAPI, Docker, Git, PyTest, Matplotlib.
- **Databases:** PL/SQL, Chroma, Pinecone and Neo4j AuraDB.
- **Cloud & DevOps:** MLOps, Kafka, Kubernetes, AWS Sagemaker/Bedrock, Private Cloud, CI/CD, HPC
- **Simulation/Design:** GEANT4 (detectors development), COMSOL (used for chips and semiconductors simulation), SolidWorks (hardware design), LaTeX

SELECTED PROJECTS

- **Subword Tokenizer + SLM Prototype** (Rust, Python, PyTorch, MPS)
 - Built a high-performance Rust BPE tokenizer and integrated it with a Python training pipeline for small language model (SLM) prototyping.
 - Trained and validated a 35.2M-parameter TinyLM on Apple Silicon (MPS) with long-run automation (3–4 hour runs) and periodic checkpointing.
 - Implemented checkpoint auto-ranking and stream retry/backoff logic, improving reliability for unattended arXiv-based training workflows.
 - [GitHub Repo](#) | [Portfolio](#)
- **Agentic Video Generator** (FastAPI, LangGraph, Redis, Postgres, ffmpeg, Streamlit, React)
 - Built an end-to-end multi-agent video generation pipeline that transforms story prompts into rendered MP4 outputs via a checkpointed LangGraph workflow.
 - Implemented asynchronous API + worker architecture with Redis queue and Postgres-backed job state for reliable long-running generation tasks.
 - Added node-level checkpoint persistence (Redis cache + Postgres durable store) with resume endpoint support to recover from mid-pipeline failures.
 - Delivered two production-ready UIs (Streamlit and React) with submission forms, status polling, resume action, and in-app video preview.
 - [GitHub Repo](#) | [Portfolio](#)
- **RAG/Monitoring System** (vector DB, retrieval, evaluation)
Details to be expanded.

PROFESSIONAL EXPERIENCE

Lead Engineer (GenAI) – Apexon (Client: AWS), Mumbai, India Dec 2025 – Present.

- Developing and leading agentic AI applications focused on autonomous reasoning, planning, and tool orchestration.
- Building multi-agent systems using OpenAI Agents SDK, CrewAI, LangGraph, AutoGen, and MCP.
- Designing production-grade agent workflows for enterprise use cases, including decision-making agents, task planners, and AI copilots.
- Collaborating with cross-functional teams to deploy scalable, secure, and responsible GenAI solutions.

Gen AI Technical Lead (IC) – FutureSoft (Client: PwC BSC), Mumbai, India Jul 2025 - Nov 2025

- Leading Generative AI projects for PwC, including enterprise-scale chatbot/agent development, RAG-based automation, and client-facing AI solutions.
- Architecting and deploying AI/ML pipelines with focus on LLMs, LangChain, and cloud-based MLOps practices.
- Mentoring cross-functional teams on GenAI adoption, responsible AI practices, and production-grade deployments.
- Collaborating with stakeholders to translate business requirements into scalable AI-driven solutions.

Postdoctoral Research Scientist (Data Scientist) – UC Davis, USA Jan 2024 – Dec 2024

- Developed AI agents for real-time data monitoring and anomaly detection using RAG, BERT, Flan-T5, and GPT.
- Integrated Pinecone vector DB for log ingestion, enabling advanced reasoning and context retention.
- Applied ML algorithms (PCA, DBSCAN, CNNs) for image classification and 3D reconstruction.
- Built Monte Carlo-based models for optimizing detector performance.

Senior Software Engineer – Nextgen Clearing Pvt Ltd., Mumbai, India Nov 2021 – May 2023

- Engineered telecom billing systems using Python, C++, and PL/SQL.
- Refactored legacy modules, boosting performance and maintainability.
- Coordinated development and debugging with cross-functional teams.

AI Research Collaborator – University of Lucknow & Fermilab, USA Mar 2019 – Oct 2021

- Developed CNN/GenAI models for particle physics image reconstruction (LArTPC).
- Optimized classification and detector performance using deep learning.
- Performed large-scale data analysis using PyTorch and Scikit-learn.

Research Fellow (Scientific Software Developer) – India-based Neutrino Observatory, Lucknow University
Aug 2014 – Mar 2019

- Created GEANT4-based simulation software for experimental physics.
- Integrated ML models to enhance detector signal recognition and efficiency.
- Deployed on Fermi Cloud (AWS/GCP-like) infrastructure for distributed computing.

EDUCATION

- **Ph.D. in Particle Physics,** University of Lucknow, India 2019 – 2022
- **M.Sc. in Physics,** Indian Institute of Technology Bombay, India 2011 – 2013
- **B.Sc. in Physics, Mathematics & Computer Science,** Ewing Christian College, Allahabad 2007 – 2010

RESEARCH PUBLICATIONS

Extensive publication record in high-energy physics and AI/ML research.
Full list available at: inspirehep.net/authors/1591917

Selected AI/ML-Enhanced Publications (Key Contributions)

- **Separation of track- and shower-like energy deposits in ProtoDUNE-SP using a convolutional neural network**, Eur. Phys. J. C 82 (2022).
Technologies: Convolutional Neural Network (CNN), supervised classification, image-based detector data.
Contributed to integration of CNN model into the reconstruction workflow and optimization with real and simulated data.
inspirehep.net/2060793
- **Highly-parallelized simulation of a pixelated LArTPC on a GPU**, JINST 18 (2023).
Technologies: GPU parallelization (CUDA), high-performance computing, simulation pipelines.
Developed and optimized GPU-accelerated simulation kernels and integrated with experiment frameworks.
inspirehep.net/2620145
- **Neutrino interaction vertex reconstruction in DUNE with integrated U-ResNet neural network** (2025).
Technologies: U-ResNet (deep residual U-Net), hit-level classification, hybrid ML pipeline integration.
Integrated U-ResNet into vertex reconstruction pipeline, ensuring efficient interfacing and runtime optimization.
inspirehep.net/2878306

CERTIFICATIONS & TEST SCORES

- AI Engineer Agentic Track: The Complete Agent and MCP Course (Udemy)
- Generative AI with Large Language Models (AWS, DeepLearning.AI)
- IIT-JAM (Physics): All India Rank – 263 (2011)
- NGPE (IAPT): Top 10 percentile (2010)

AWARDS & FELLOWSHIPS

- Rajendra Raja Fellowship – Fermilab, USA
- INO Fellowship – DST, India

EXTRACURRICULAR ACTIVITIES

- Winner: 4x100m Relay (IIT-Bombay & ECC), Basketball/Volleyball tournaments
- Finalist: 400m Sprint, Annual college meets (2008–2010)